

Roll No. : .....

**MID SEMESTER EXAMINATION SEPT. 2025**

Name of the Program: **B.Tech CSE (AI/ML Specialization)**

Semester: **5th**

Name of the Course : Natural Language Processing and Computer Vision

Course Code: TCS 564

Time : 90 Minutes

Maximum Marks : 50

**Note:**

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

|           |                                                                                                                                                                                                                                                             |            |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Q1</b> | (10 Marks)                                                                                                                                                                                                                                                  | <b>CO1</b> |
| (a)       | What are the different challenges we encounter while handling Natural Languages compared to formal languages? Briefly discuss some application areas of NLP.                                                                                                |            |
|           | OR                                                                                                                                                                                                                                                          |            |
| (b)       | "NLP=NLU+NLG" – Explain this. Can you perform Chunking without doing POS tagging? Give reasons to support your answer.                                                                                                                                      |            |
| <b>Q2</b> | (10 Marks)                                                                                                                                                                                                                                                  | <b>CO2</b> |
| (a)       | Differentiate between Inflectional and Derivational forms of words. What differentiates Stemming from Lemmatization?                                                                                                                                        |            |
|           | OR                                                                                                                                                                                                                                                          |            |
| (b)       | "Overstemming leads to loss of precision and Understemming leads to loss of recall." Explain this using suitable example.                                                                                                                                   |            |
| <b>Q3</b> | (10 Marks)                                                                                                                                                                                                                                                  | <b>CO2</b> |
| (a)       | Why do we perform POS tagging in NLP? Differentiate between Open class and Closed class in POS tagging. Write code in spaCy to show POS tagging for both coarse grained and fine grained cases.                                                             |            |
|           | OR                                                                                                                                                                                                                                                          |            |
| (b)       | Discuss three use cases of Named Entity Recognition. Write code in spaCy to perform NER on a given sentence. Also use displacy visualization for the same.                                                                                                  |            |
| <b>Q4</b> | (10 Marks)                                                                                                                                                                                                                                                  | <b>CO1</b> |
| (a)       | You are given the three following sentences:<br>Sun shines brightly.<br>Stars twinkle at night.<br>Moon is dull today.<br>(i) Create a Bag of Words matrix for the above. (ii) Write code (either using NLTK or spaCy) for the same.                        |            |
|           | OR                                                                                                                                                                                                                                                          |            |
| (b)       | You are given the three following sentences:<br>The cat sat on the mat.<br>The dog sat on the log.<br>The cat chased the dog.<br>(i) Calculate the term frequency of each word in each sentence. (ii) Calculate the Inverse Document Frequency of each word |            |
| <b>Q5</b> | (10 Marks)                                                                                                                                                                                                                                                  | <b>CO1</b> |
| (a)       | What are ngrams? Explain how will you use ngrams to predict words (use bigrams as example). Show how Markov assumption helps in the chain rule of probability.                                                                                              |            |
|           | OR                                                                                                                                                                                                                                                          |            |
| (b)       | Explain how GloVe uses the word co-occurrence matrix to learn word embeddings. Compare GloVe with Word2Vec in terms of how they capture word semantics.                                                                                                     | <b>CO2</b> |

